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SECOND REVIEW UNDER THE STAFF-MONITORED PROGRAM—DEBT SUSTAINABILITY ANALYSIS

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Myanmar is assessed to be at low risk of debt distress following the clearance of its external arrears.[1,2] The baseline scenario assumes the planned resolution of Myanmar's external arrears will be completed in 2013/14 as agreed with the Paris Club creditors. The baseline scenario also assumes a gradual reduction in Myanmar's reliance on nonconcessional financing, as more concessional resources become available after the resolution of arrears. Under the baseline scenario, public and publicly guaranteed (PPG) external debt burden indicators remain well below their indicative thresholds. Similarly, under the baseline scenario total public debt will also be below the benchmark and will continue to decline over the medium and long term. However, shocks lead to the indicative benchmark for the present value of total public debt being exceeded. While Myanmar's risk of debt distress can be characterized as low, it requires close monitoring, in particular because of the relatively high level of domestic debt. Maintaining the low-risk categorization will require continuation of the current prudent fiscal policy, improvements in tax policy and public financial management and increasing use of concessional finance while limiting nonconcessional borrowing to only viable and growth-enhancing projects.

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[1] External public and publicly guaranteed (PPG) debt and public domestic debt dynamics are assessed using the LICDSA framework, which recognizes that better policies and institutions allow countries to manage higher levels of debt, and thus the threshold levels are policy-dependent. The quality of a country's policies and institutions are normally measured by the World Bank's Country Policy and Institutional Assessment (CPIA). Since Myanmar does not currently have a CPIA rating, the most conservative thresholds are applied for the purposes of this DSA. The current DSA framework uses the 5 percent discount rate specified in the new guidance note.

[2] The DSA was jointly prepared by the IMF and the World Bank in consultation with the Asian Development Bank.

BACKGROUND

1. The external and public debt sustainability analyses are based on the standard LIC DSA framework. The DSA framework presents the projected path of Myanmar's external and public sector debt burden indicators, and draws some conclusions on the forward-looking sustainability of debt.

2. Myanmar's DSA builds on revised macroeconomic assumptions presented in Box 1. Compared to the previous DSA, the assumptions take into account recent developments. In particular, the medium-term GDP growth rate has been revised upward to 7.8 percent from an average of about 7 percent in the previous DSA reflecting robust FDI and related investment in productive sectors, as well as in public investment.[3] The growth outlook has strengthened due to better than expected performance of the services sector and construction which is forecast to continue over the medium-term. In addition, special export zones are expected to start operations which should boost manufacturing and lead to robust growth over the medium term. The overall fiscal deficit remains the same as in the previous DSA in the medium term, although revenue collection is expected to improve. Long-term growth and inflation, however, are expected to be approximately the same as in the previous DSA.

3. The current DSA incorporates the agreed resolution of Myanmar's arrears with its multilateral and bilateral creditors. In April 2012, Myanmar agreed with Japan, its largest creditor, on a debt restructuring plan to resolve its arrears through a bridge loan operation and cancelation of principal payments and overdue charges.[4] Furthermore, Myanmar resolved its arrears to the World Bank and the Asian Development Bank in January 2013, with the help of a bridge financing operation from Japan. In late-January 2013, the Paris Club reached an agreement with the Myanmar authorities on a debt treatment to be completed in early 2014. Paris Club members agreed to write off 50% of all arrears and reschedule the remaining arrears over 15 years with a 7 year grace period. The treatment will be phased, with 25% of the written-off occurring immediately and 75% on a successful completion of the SMP. Japan signed the minute but will continue with the more generous treatment that it had previously announced. Norway and Denmark, also provided a more generous treatment by writing off 100% of their outstanding debt. Similarly, Italy signed a bi-lateral agreement with Myanmar to cancel 50% percent of all arrears and conduct a debt-for development swap of the remaining arrears. As of end-January 2014, nine of the eleven Paris Club creditors had signed bilateral agreements with Myanmar and the signing of the remaining two agreements is imminent.

4. Myanmar is expected to gradually reduce its reliance on external nonconcessional financing. With the expected resolution of its arrears and the re-engagement of the international community,

[3] The staff of the World Bank and the IMF, while strong supporters of the overall reform effort, reserve some caution given the newness of the reform program.

[4] In FY2012/13 Japan cancelled arrears on payments due after 2003. Principal and interest arrears on pre-2003 payments due were rescheduled on very concessional terms also before the end of FY2012/13. Before the end of FY2013/14, Japan also cancelled late interest charges on pre-2003 arrears. The implementation of this agreement was done in the context of the Paris Club framework.

Box 1. Myanmar: Key Macroeconomic Assumptions Underlying the DSA for the Baseline Scenario (FY2013/14–33/34)

- Real GDP growth rate is assumed to average about 7.8 percent over the medium term driven by commodity exports and higher investment due to the implementation of Myanmar's structural reform plans, with the support of the international community, and suspension of sanctions. Over the long term, real GDP growth rate is expected to average 7.4 percent supported by prudent macroeconomic policies and robust FDI and related investments.
- Inflation is projected on average at 6.2 percent (y/y) in the medium term and expected to continue trending down and reach around 3 percent in FY2033/34. This is based on the assumption that deficit monetization will be replaced with bond financing, as well as having a consistent monetary policy framework in place.
- The primary fiscal deficit is expected to average at 2.8 percent of GDP over the medium term. Revenues are expected to increase significantly over the medium term, primarily due to a substantial increase in gas export revenues expected with the completion of Shwe and Zawtika projects as well as planned improvements to tax administration and efforts to broaden the tax bases. Expenditures are also expected to increase, reflecting Myanmar's significant development needs.
- Exports as a percentage of GDP is assumed to average around 27.5 percent, in the medium term reflecting mainly new natural gas projects coming on stream. Over the longer term, export growth remains robust, averaging around 34.5 percent during FY2019/20–FY2033/34, driven by export diversification beyond commodity exports supported by higher investment, including foreign direct investment.
- Imports as a percentage of GDP is assumed to average around 30.9 percent in the medium term given the authorities' plan to remove exchange restrictions on current international payments and transfers and the import content of higher investment including FDI. Over the longer term, imports as a percentage of GDP will average around 39.6 percent during FY2019/20–FY2033/34, reflecting the need for imported capital goods to support growth.
- The noninterest current account (including official grants), is expected to remain in deficit in the medium term at an average of 4.5 percent of GDP. Over the long term, the noninterest current account deficit is expected to stabilize at around 4.6 percent of GDP.
- As of end-FY2012/13, around 35 percent of public domestic debt was in the form of treasury bonds, with the rest in the form of treasury bills (bearing a nominal interest rate of 4 percent). Treasury bonds are issued in maturities of two, three, and five years, bearing nominal interest at fixed interest rates of 8.75, 9, and 9.5 percent, respectively. Average real interest rates on domestic public debt turned positive in FY2011/12 as inflation subsided but are projected to narrow in FY2013/14, due to rising inflation and fixed nominal interest rates. Real interest rates are expected remain positive in the short term, in line with government's plan to further shift toward bond financing of the deficit and moderate inflation, and widen in the medium term, as the government moves toward a market based deficit financing.
- Financing of the deficit is expected to be met by a combination of domestic public borrowing and foreign loans, including from bilateral and multilateral creditors. In the short term, deficit financing will rely on non-concessional external financing and increasingly on the issuance of Treasury bonds (instead of Treasury bills in order to reduce deficit monetization). Over the medium term, the share of external financing is expected to increase, especially from concessional sources. Nonetheless, Treasury bond issuance is expected to continue over the medium term (despite the higher cost relative to external financing) consistent with the goal to develop a domestic Treasury bond market. This will also facilitate the liberalization of interest rates with the planned move to treasury security auctions.

Myanmar is expected to gradually regain access to concessional resources. As donors re-engage with Myanmar, and gradually identify suitable projects, the share of nonconcessional financing is expected to decline. The authorities aim to use nonconcessional external borrowing only to finance economically viable projects in priority sectors such as energy and infrastructure, at levels consistent with low risk of debt distress. In order to limit the use of nonconcessional financing, the authorities also aim to keep the fiscal deficit in FY2013/14 as well as over the medium term at 5 percent of GDP.

Key Macroeconomic Assumptions Underlying the DSA for the Baseline Scenario (FY2013/14–33/34)

	DSA Update Baseline		2013 DSA Baseline	
	2013/14–2018/19	2019/20–2033/34	2013/14–2018/19	2019/20–2033/34
Real GDP Growth (in percent)	7.8	7.4	7.0	7.4
Inflation (in percent)	6.2	3.8	5.6	3.8
Overall fiscal balance (in percent of GDP)	-4.8	-4.2	-4.9	-4.2
Noninterest current account (in percent of GDP)	-4.5	-4.6	-4.2	-4.7
Revenue (nonfinancial public sector; in percent of GDP)	23	24.3	24.7	26.7

Source: IMF staff estimates

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

5. Successful completion of the resolution of the remaining arrears in 2013/14 will ensure that all external debt indicators are below their indicative thresholds in 2013/14. Under the baseline scenario, which assumes the completion of the second phase of the resolution of Myanmar's arrears, all external debt indicators will be below their indicative thresholds at end-FY2013/14. The impact of the debt rescheduling is evident in the decline in the debt burden indicators between FY2012/13 and FY2013/14 (Table 1a). For example, the PV of the debt-to-GDP ratio decreases from 18.9 percent in FY2012/13 to 12.8 percent in FY2013/14.

Myanmar: External PPG Debt Indicators

	Indicative Thresholds	End 2012/13	End 2013/14
NPV of debt as a percent of:			
GDP	30	18.9	12.8
Exports	100	91.6	55.2
Revenue	200	81.1	58.4
Debt Service, as a percentage of:			
Exports	15	2.2	4.1
Revenue	18	2.0	4.4

Sources: IMF staff estimates

6. Standard stress tests indicate that Myanmar's external debt sustainability may be vulnerable to export shocks, but remains in a low risk of debt distress. Although no thresholds are breached, the export shock leads external debt to come close to the PV of debt-to-export indicative threshold temporarily (Figure 1 and Table 1b).[5] However, this shock is introduced at the time of the projected increase in exports due to the completion of the Shwe and Zawtika gas projects. Myanmar's good track record of

[5] The typical historical scenario is not shown in this analysis. In the case of Myanmar, the historical scenario would imply an unlikely return to pre-reform policies: low noninterest current account deficits (consistent with binding international sanctions) and sustained real exchange rate pressures.

implementing natural gas sector projects (e.g. the Yadana and Yetagun projects are fully operational) and contractual purchasing commitments by project partners (projects partners have agreed to purchase most of the production from these projects), mitigate some of the risks of this particular shock i.e. a significant slowdown in exports compared to the baseline scenario.[6] Accordingly, Myanmar can be classified as at low risk of debt distress.

PUBLIC DEBT SUSTAINABILITY ANALYSIS

7. Public debt burden indicators (including domestic public debt) are expected to improve initially, reflecting the impact of the planned external arrears resolution. The nominal debt stock would decrease from 46.3 percent in 2012/13 to 41.1 percent in 2013/14 mainly due to the clearance of the arrears. Thereafter, the public debt is expected to slightly decline (Figure 2 and Table 1c). This reflects the significant development needs of Myanmar and the associated overall fiscal deficits assumed in the baseline scenario.

8. Although total public sector debt is relatively high, it remains below the indicative benchmark under the baseline scenario. The PV of total public debt as a percentage of GDP remains below the indicative benchmark and continues to fall over the medium to long-term. This trajectory of total public debt underscores the need to limit the share of nonconcessional financing as is assumed in the baseline scenario. It should also be noted that this level of debt can give rise to vulnerabilities including the burden of high debt service. It is also susceptible to shocks (see the paragraph below for implications of shocks). Risks are somewhat mitigated as the authorities aim to use their borrowing to only finance economically viable projects in priority sectors, which will be crucial to growth.

9. Stress tests indicate the presence of vulnerabilities. In particular, public debt sustainability is vulnerable to lower real GDP growth, and large and persistent primary fiscal deficits (Table 1d). The stabilization of the debt-to-GDP ratio at a relatively high level under the baseline scenario and the vulnerabilities indicated by the stress tests highlight the need for a prudent fiscal policy, through improvements in tax policy and public financial management including strengthening of debt management functions, and project assessment (i.e. ability to identify projects with good economic returns) and capacity to obtain more concessional financing.

STAFF ASSESSMENT

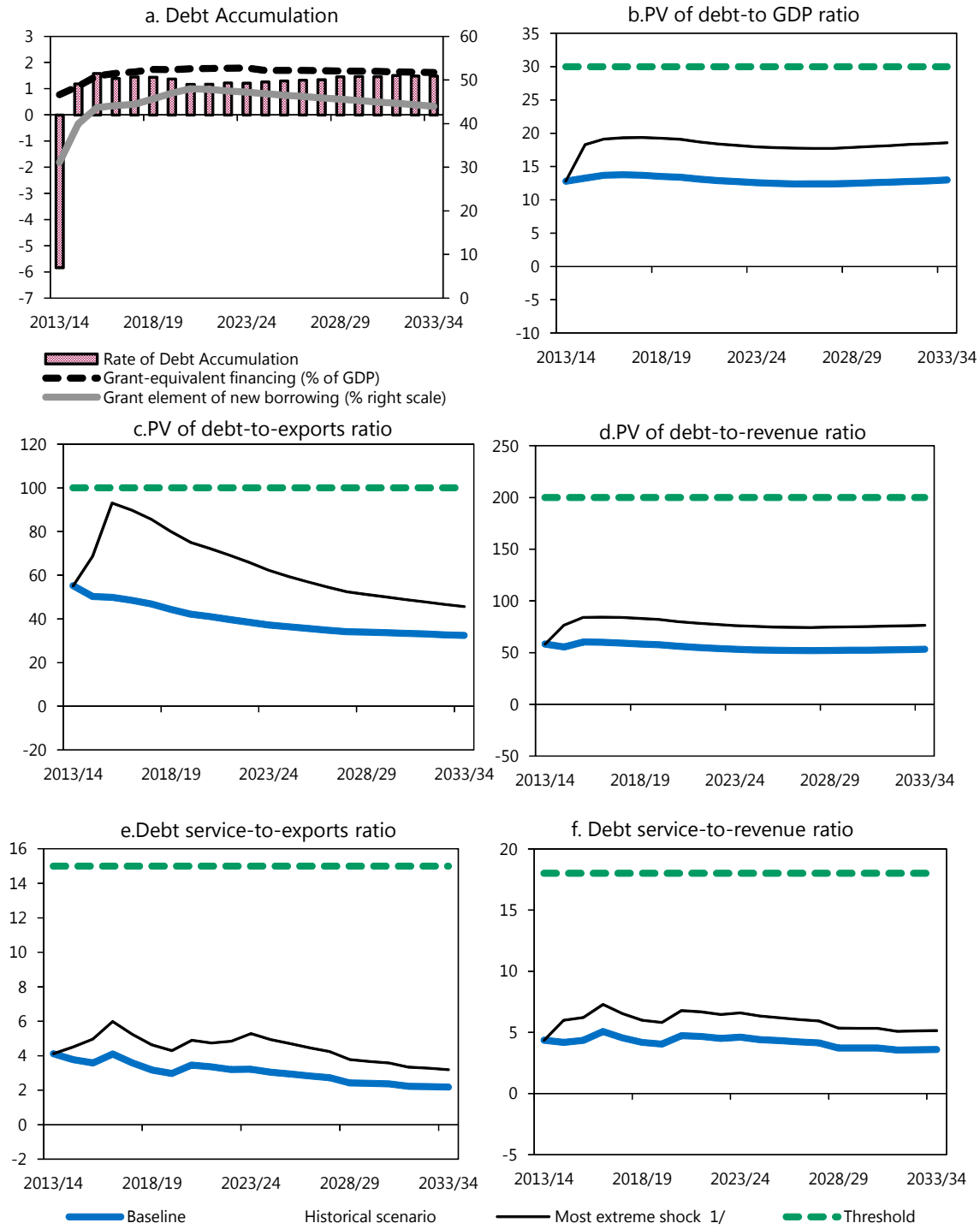
10. Myanmar is assessed to be at low risk of debt distress given its planned resolution of external arrears. Under the baseline scenario, which assumes a resolution of the remaining arrears in 2013/14 as agreed with the Paris Club, all indicators are below their indicative thresholds with the PV of debt-to-export ratio temporarily coming close to the threshold under the export shock. As explained above, the risks associated with this shock are mitigated by the good track record of Myanmar in terms of

[6] There is a growing likelihood that decreases in global gas prices could accelerate as a result of market reaction to upcoming U.S. exports. However, the possible drop in prices in mitigated by the potential increase in the volume of gas exports as exploration of gas fields in a significant number of block will start at the end of this year.

implementation of large natural gas projects as well as contractual purchase commitments of project partners. However, it also underscores the importance of diversifying Myanmar's export base. Under the baseline scenario, the total public debt remains below the benchmark and continues to fall over the projection period. But caution must be exercised as stress test shows that the benchmark can be breached. Overall public debt sustainability is vulnerable to fiscal slippages and low real GDP growth. This highlights the need for prudent fiscal policy, as well as improvements in tax policy to reduce reliance on nontax revenues from nonrenewable sources and public financial and debt management.

11. The authorities broadly agreed with these conclusions and with the thrust of the analysis. They concurred with staff on the need to be cautious on nonconcessional borrowing. They acknowledged the importance of maintaining high exports and also financing only viable projects that would enhance growth.

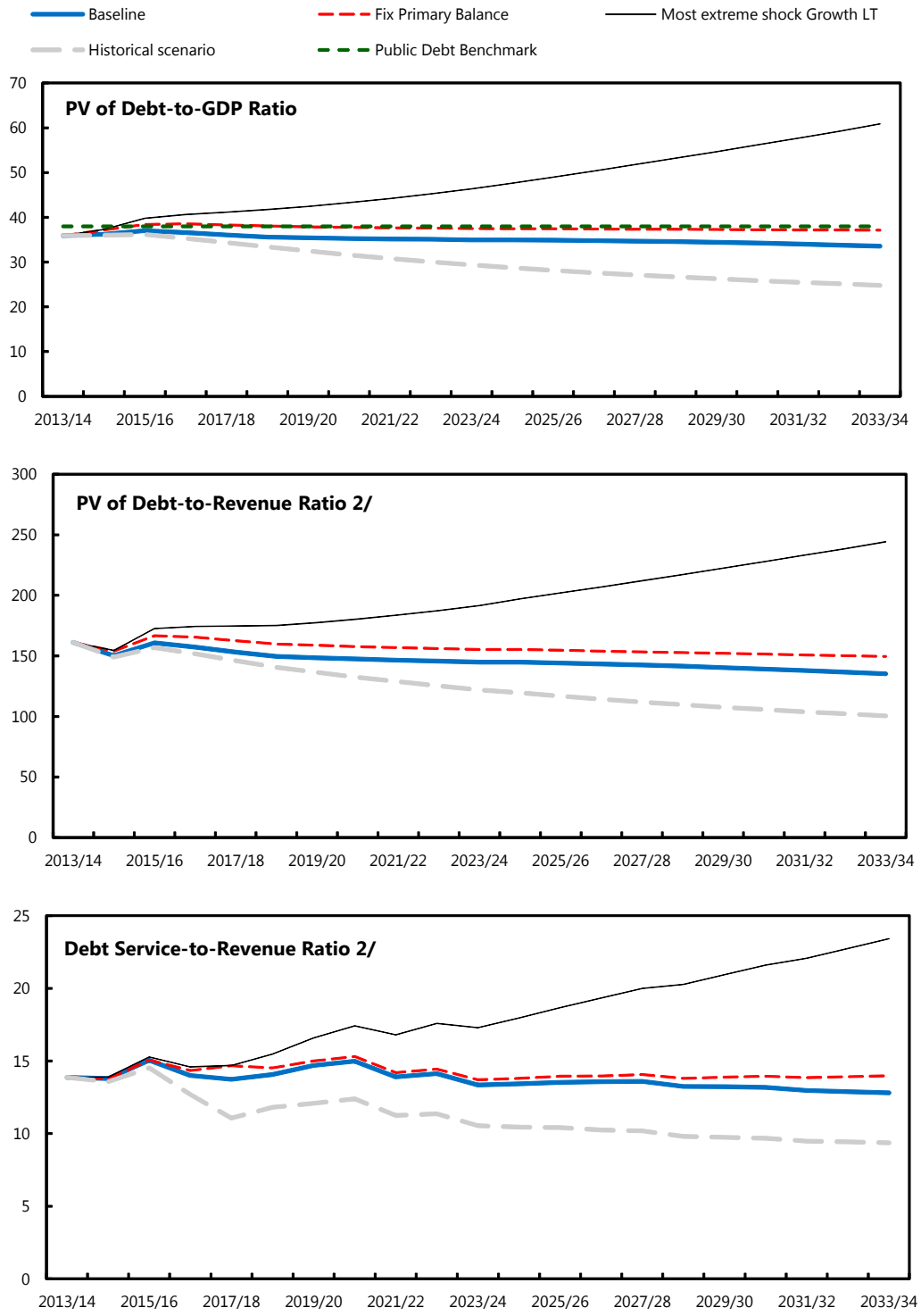
Figure 1. Myanmar: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013/14–2033/34 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a One-time depreciation shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Myanmar: Indicators of Public Debt Under Alternative Scenarios, 2013/14–2033/34 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023.
2/ Revenues are defined inclusive of grants.

Table 1a. Myanmar: External Debt Sustainability Framework, Baseline Scenario, 2010/11–2033/34 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 2/ Average	Standard 2/ Deviation	Projections									
												2013/14–2018/19			2019/20–2033/34
	2010/11	2011/12	2012/13			2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	Average	2023/24	2033/34	Average
External debt (nominal) 1/	**29.0**	**27.3**	**24.2**			**18.3**	**18.7**	**19.8**	**20.3**	**20.5**	**20.5**		**20.2**	**21.2**	
of which: Public and publicly guaranteed (PPG)	29.0	27.3	24.2			18.3	18.7	19.8	20.3	20.5	20.5		20.2	21.2	
Change in external debt	-7.3	-1.6	-3.1			-5.9	0.4	1.1	0.5	0.2	0.1		0.0	0.1	
Identified net debt-creating flows	-11.5	-4.9	-0.5			-1.3	-0.3	-1.2	-1.6	-1.7	-1.7		-2.1	-1.6	
Non-interest current account deficit	**0.9**	**1.1**	**4.2**	**-1.1**	**4.1**	**4.5**	**4.7**	**4.4**	**4.4**	**4.3**	**4.4**		**4.5**	**4.6**	4.6
Deficit in balance of goods and services	-0.3	1.1	3.9			3.8	2.9	3.1	3.2	3.1	4.2		4.8	4.8	
Exports	18.7	19.6	20.6			23.2	26.4	27.5	28.3	29.2	30.5		33.7	40.0	
Imports	18.4	20.7	24.5			27.0	29.3	30.6	31.5	32.3	34.7		38.5	44.8	
Net current transfers (negative = inflow)	-0.5	-0.9	-0.9	-0.9	0.2	-1.0	-1.1	-1.2	-1.3	-1.4	-1.5		-1.4	-1.1	-1.2
of which: Official	-0.1	-0.1	-0.1			-0.1	-0.2	-0.3	-0.4	-0.5	-0.6		-0.6	-0.5	
Other current account flows (negative = net inflow)	1.7	0.8	1.2			1.7	3.0	2.5	2.6	2.6	1.7		1.1	0.9	
Net FDI (negative = inflow)	**-4.5**	**-3.7**	**-5.0**	**-3.0**	**1.1**	**-4.4**	**-4.1**	**-4.6**	**-4.9**	**-4.9**	**-5.0**		**-5.5**	**-5.0**	-5.4
Endogenous debt dynamics 3/	**-7.8**	**-2.4**	**0.4**			**-1.5**	**-1.0**	**-1.0**	**-1.1**	**-1.1**	**-1.1**		**-1.1**	**-1.2**	
Contribution from nominal interest rate	0.6	1.0	0.2			0.3	0.3	0.3	0.3	0.3	0.3		0.3	0.3	
Contribution from real GDP growth	-1.5	-1.5	-2.0			-1.8	-1.3	-1.4	-1.4	-1.4	-1.4		-1.4	-1.4	
Contribution from price and exchange rate changes	-7.0	-1.9	2.2			...	...	...	...	...	...		...	...	
Residual (3-4) 4/	**4.2**	**3.3**	**-2.7**			**-4.6**	**0.8**	**2.2**	**2.1**	**1.9**	**1.8**		**2.0**	**1.7**	
of which: Resolution of arrears	0.0	0.0	-10.9			-8.5	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 5/	...	...	18.9			12.8	13.3	13.7	13.8	13.7	13.5		12.6	13.0	
In percent of exports	...	...	91.6			55.2	50.3	49.9	48.6	46.8	44.3		37.3	32.4	
PV of PPG external debt	**...**	**...**	**18.9**			**12.8**	**13.3**	**13.7**	**13.8**	**13.7**	**13.5**		**12.6**	**13.0**	
In percent of exports	**...**	**...**	**91.6**			**55.2**	**50.3**	**49.9**	**48.6**	**46.8**	**44.3**		**37.3**	**32.4**	
In percent of government revenues	**...**	**...**	**81.1**			**58.4**	**55.7**	**60.3**	**60.1**	**59.4**	**58.3**		**53.4**	**53.4**	
Debt service-to-exports ratio (in percent)	**7.5**	**10.6**	**2.2**			**4.1**	**3.8**	**3.6**	**4.1**	**3.6**	**3.2**		**3.2**	**2.2**	
PPG debt service-to-exports ratio (in percent)	**7.5**	**10.6**	**2.2**			**4.1**	**3.8**	**3.6**	**4.1**	**3.6**	**3.2**		**3.2**	**2.2**	
PPG debt service-to-revenue ratio (in percent)	**12.2**	**17.2**	**2.0**			**4.4**	**4.2**	**4.3**	**5.1**	**4.6**	**4.2**		**4.6**	**3.6**	
Total gross financing need (Billions of U.S. dollars)	-0.9	-0.1	-0.1			0.8	1.2	0.8	0.6	0.6	0.6		0.5	2.7	
Non-interest current account deficit that stabilizes debt ratio	8.2	2.7	7.3			10.4	4.3	3.4	3.9	4.1	4.4		4.5	4.5	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.3	5.9	7.3	9.3	4.2	7.5	7.8	7.8	7.8	7.8	7.8	7.8	7.4	7.4	7.4
GDP deflator in U.S. dollar terms (change in percent)	23.8	6.9	-7.5	12.7	18.2	-5.9	-0.8	-0.3	1.4	2.6	3.2	0.0	3.3	3.0	3.2
Effective interest rate (percent) 6/	2.2	3.9	0.6	1.7	1.0	1.3	1.8	1.8	1.8	1.7	1.7	1.7	1.5	1.4	1.4
Growth of exports of G&S (U.S. dollar terms, in percent)	16.1	18.6	4.5	14.9	13.3	14.1	21.4	12.0	12.5	14.1	16.2	15.1	13.1	12.9	12.9
Growth of imports of G&S (U.S. dollar terms, in percent)	18.6	27.7	17.6	19.7	23.9	11.3	15.9	12.5	12.3	13.4	19.7	14.2	12.9	12.4	12.8
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	31.1	40.0	43.7	44.1	44.4	45.7	41.5	47.2	44.0	46.1
Government revenues (excluding grants, in percent of GDP)	11.4	12.0	23.3			21.9	23.8	22.7	22.9	23.0	23.2		23.6	24.3	23.8
Aid flows (in Billions of U.S. dollars) 7/	0.0	0.0	0.0			0.2	0.6	1.0	1.2	1.4	1.8		3.3	7.7	
of which: Grants	0.0	0.0	0.0			0.2	0.2	0.2	0.3	0.4	0.5		0.9	2.1	
of which: Concessional loans	0.0	0.0	0.0			0.0	0.4	0.8	0.9	1.0	1.2		2.4	5.7	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			0.8	1.1	1.5	1.6	1.6	1.7		1.8	1.6	1.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			44.0	47.8	49.3	51.4	53.5	56.3		57.3	53.2	55.5
Memorandum items:															
Nominal GDP (Billions of U.S. dollars)	49.6	56.2	55.8			56.4	60.3	64.8	70.8	78.3	87.1		146.9	412.3	
Nominal dollar GDP growth	30.4	13.2	-0.7			1.2	6.9	7.5	9.3	10.6	11.3	7.8	10.9	10.6	10.9
PV of PPG external debt (in Billions of U.S. dollars)			10.3			7.0	7.7	8.6	9.5	10.6	11.7		18.4	53.4	
(PVt-PVt-1)/GDPt-1 (in percent)						-5.8	1.2	1.6	1.4	1.5	1.4	0.2	1.2	1.5	1.4
Gross workers' remittances (Billions of U.S. dollars)	0.2	0.4	0.5			0.5	0.6	0.6	0.7	0.7	0.8		1.1	2.4	
PV of PPG external debt (in percent of GDP + remittances)	...	...	18.7			12.7	13.1	13.6	13.6	13.6	13.4		12.5	12.9	
PV of PPG external debt (in percent of exports + remittances)	...	...	87.9			53.1	48.6	48.2	47.0	45.4	43.1		36.4	32.0	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.1			4.0	3.6	3.5	4.0	3.5	3.1		3.1	2.1	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
3/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
4/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
5/ Assumes that PV of private sector debt is equivalent to its face value.
6/ Current-year interest payments divided by previous period debt stock.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Myanmar: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013/14–2033/34

(In percent)

	Projections							
	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	**2023/24**	2033/34
PV of debt-to GDP ratio								
Baseline	13	13	14	14	14	14	**13**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013–2033 1/	13	8	6	4	2	1	**-3**	-5
A2. New public sector loans on less favorable terms in 2013–2033 2/	13	13	15	15	16	16	**17**	20
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014–2015	13	13	14	14	14	14	**13**	14
B2. Export value growth at historical average minus one standard deviation in 2014–2015 3/	13	15	19	19	19	19	**16**	14
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2014–2015	13	13	15	15	15	15	**14**	14
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014–2015 4/	13	14	16	16	16	16	**14**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	13	15	18	18	18	17	**15**	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	13	18	19	19	19	19	**18**	19
PV of debt-to-exports ratio								
Baseline	55	50	50	49	47	44	**37**	32
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013–2033 1/	55	32	21	14	7	2	**-9**	-14
A2. New public sector loans on less favorable terms in 2013–2033 2/	55	50	53	54	55	54	**52**	51
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014–2015	55	48	48	48	46	44	**37**	32
B2. Export value growth at historical average minus one standard deviation in 2014–2015 3/	55	69	93	90	86	80	**62**	46
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2014–2015	55	48	48	48	46	44	**37**	32
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014–2015 4/	55	54	60	58	56	53	**42**	34
B5. Combination of B1-B4 using one-half standard deviation shocks	55	65	81	78	75	70	**55**	40
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	55	48	48	48	46	44	**37**	32
PV of debt-to-revenue ratio								
Baseline	58	56	60	60	59	58	**53**	53
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013–2033 1/	58	35	26	17	9	3	**-13**	-22
A2. New public sector loans on less favorable terms in 2013–2033 2/	58	56	64	67	69	71	**74**	84
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014–2015	58	55	62	62	62	61	**56**	56
B2. Export value growth at historical average minus one standard deviation in 2014–2015 3/	58	64	85	84	82	80	**68**	57
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2014–2015	58	56	65	65	65	64	**59**	59
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014–2015 4/	58	59	72	72	71	69	**61**	55
B5. Combination of B1-B4 using one-half standard deviation shocks	58	62	79	78	76	74	**63**	53
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	58	77	84	84	84	83	**76**	76

Table 1b. Myanmar: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013/14–2033/34 (concluded)

(In percent)

	Projections							
	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	**2023/24**	2033/34
Debt service-to-exports ratio								
Baseline	4	4	4	4	4	3	**3**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013–2033 1/	4	3	3	2	2	1	**1**	0
A2. New public sector loans on less favorable terms in 2013–2033 2/	4	4	4	4	4	4	**4**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014–2015	4	4	4	4	4	3	**3**	2
B2. Export value growth at historical average minus one standard deviation in 2014–2015 3/	4	5	5	6	5	5	**5**	3
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2014–2015	4	4	4	4	4	3	**3**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014–2015 4/	4	4	4	4	4	3	**4**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	4	5	5	4	**5**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	4	4	4	4	3	**3**	2
Debt service-to-revenue ratio								
Baseline	4	4	4	5	5	4	**5**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013–2033 1/	4	4	3	3	2	2	**1**	0
A2. New public sector loans on less favorable terms in 2013–2033 2/	4	4	4	5	5	5	**6**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014–2015	4	4	5	5	5	4	**5**	4
B2. Export value growth at historical average minus one standard deviation in 2014–2015 3/	4	4	5	6	5	5	**6**	4
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2014–2015	4	4	5	6	5	5	**5**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014–2015 4/	4	4	4	5	5	4	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	4	5	5	4	**5**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	6	6	7	7	6	**7**	5
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	44	44	44	44	44	44	**44**	44

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 1c. Myanmar: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010/11–2033/34

(In percent of GDP, unless otherwise indicated)

	Actual					Projections									
	2010/11	2011/12	2012/13	Average 1/	Standard Deviation 1/	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2013/14–18/19 Avg.	2023/24	2033/34	2019/20–33/34 Avg.
Public sector debt 2/	50.3	50.3	46.1			40.7	40.8	42.5	42.7	42.5	42.4		42.5	41.8	
of which: Foreign-currency denominated	29.0	27.3	24.2			18.3	18.7	19.8	20.3	20.5	20.5		20.2	21.2	
Change in public sector debt	-4.6	0.0	-4.2			-5.5	0.1	1.7	0.2	-0.1	-0.1		0.0	-0.3	
Identified debt-creating flows	-7.0	-1.4	2.1			1.0	-0.5	0.5	-0.6	-0.8	-0.5		-0.1	-0.2	
Primary deficit	4.2	2.9	2.0	2.4	1.0	2.6	1.9	3.4	2.8	2.9	3.0	2.8	3.0	2.5	2.9
Revenue and grants 3/	11.4	12.0	23.3			22.3	24.1	23.0	23.3	23.5	23.8		24.2	24.8	
of which: Grants	0.0	0.0	0.1			0.3	0.3	0.3	0.4	0.5	0.6		0.6	0.5	
Primary (noninterest) expenditure 3/	15.6	14.9	25.3			24.9	26.0	26.5	26.1	26.5	26.8		27.1	27.4	
Automatic debt dynamics	-9.8	-3.2	0.7			-1.3	-2.4	-2.9	-3.5	-3.7	-3.6		-3.1	-2.8	
Contribution from interest rate/growth differential	-3.5	-1.9	-3.2			-3.3	-3.0	-3.1	-3.4	-3.4	-3.2		-2.8	-2.3	
of which: Contribution from average real interest rate	-0.7	0.9	0.3			-0.1	-0.1	-0.1	-0.4	-0.3	-0.1		0.1	0.7	
of which: Contribution from real GDP growth	-2.8	-2.8	-3.4			-3.2	-2.9	-3.0	-3.1	-3.1	-3.1		-2.9	-2.9	
Contribution from real exchange rate depreciation	-6.2	-1.3	3.8			2.0	0.6	0.2	0.0	-0.3	-0.4		...	...	
Other identified debt-creating flows	-1.4	-1.1	-0.5			-0.4	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-1.4	-1.1	-0.5			-0.4	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	2.3	1.4	-6.3			-6.4	0.7	1.2	0.8	0.6	0.4		0.1	-0.1	
Other Sustainability Indicators															
PV of public sector debt	...	...	41.4			35.9	36.2	37.0	36.6	36.0	35.6		35.0	33.6	
of which: Foreign-currency denominated	...	...	18.9			12.8	13.3	13.7	13.8	13.7	13.5		12.6	13.0	
of which: External	...	...	18.9			12.8	13.3	13.7	13.8	13.7	13.5		12.6	13.0	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 4/	7.2	6.7	5.3			6.7	6.6	7.9	6.9	6.8	6.8		6.6	5.9	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	177.4			161.2	150.1	160.8	157.2	153.1	149.6		144.8	135.3	
PV of public sector debt-to-revenue ratio (in percent)	...	...	177.8			163.5	151.9	162.8	159.9	156.4	153.4		148.5	138.1	
of which: External 5/	...	...	81.1			58.4	55.6	60.3	60.1	59.3	58.3		53.3	53.4	
Debt service-to-revenue and grants ratio (in percent) 6/	24.8	32.1	10.3			13.9	13.8	15.0	14.0	13.7	14.1		13.4	12.8	
Debt service-to-revenue ratio (in percent) 6/	24.8	32.1	10.3			14.1	14.0	15.2	14.2	14.0	14.4		13.7	13.1	
Primary deficit that stabilizes the debt-to-GDP ratio	8.8	2.9	6.2			8.1	1.7	1.7	2.7	3.0	3.1		3.0	2.8	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	5.3	5.9	7.3	9.3	4.2	7.5	7.8	7.8	7.8	7.8	7.8	7.8	7.4	7.4	7.4
Average nominal interest rate on forex debt (in percent)	2.3	4.0	0.6	1.7	1.0	1.3	1.9	1.8	1.8	1.8	1.7	1.7	1.5	1.4	1.5
Average nominal interest rate on domestic debt (in percent)	5.5	5.6	5.3	4.9	0.5	6.3	6.2	6.2	4.7	4.5	4.8	5.4	5.0	5.0	5.0
Average real interest rate (in percent)	-1.4	2.0	0.6	-1.3	1.7	-0.1	-0.2	-0.3	-0.9	-0.9	-0.3	-0.5	0.2	1.7	0.4
Average real interest rate on foreign-currency debt (in percent)	-3.0	-2.4	-1.5	-2.2	0.6	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	-1.8	0.1	-1.8
Average real interest rate on domestic debt (in percent)	-2.5	2.8	2.4	-5.9	7.4	0.4	-0.4	-0.6	-1.7	-1.5	-0.5	-0.7	0.8	2.0	1.2
Real exchange rate depreciation (in percent, + indicates depreciation)	-18.3	-4.7	15.2	-6.1	14.0	9.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.2	2.8	2.8	12.1	8.5	5.8	6.6	6.9	6.5	6.1	5.3	6.2	4.2	3.0	3.8
Growth of real primary spending (deflated by GDP deflator, in percent) 3/	14.6	0.2	91.4	20.8	27.4	6.4	13.8	8.4	5.6	8.4	8.5	8.5	7.5	7.4	7.5
Grant element of new external borrowing (in percent)	...	...	...	...	...	31.1	40.0	43.7	44.1	44.4	45.7	41.5	47.2	44.0	...

Sources: Country authorities; and staff estimates and projections.

1/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

2/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

3/ The valuation effect raised revenues and expenditures in FY2012/13 due to the exchange rate unification on April 1, 2012.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

5/ Revenues excluding grants.

6/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

Table 1d. Myanmar: Sensitivity Analysis for Key Indicators of Public Debt 2013/14–2033/34

	Projections							
	2013/14	2014/15	2015/16	2016/17	2017/18	2018/19	2023/24	2033/34
PV of Debt-to-GDP Ratio								
Baseline	36	36	37	37	36	36	35	34
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	36	36	36	35	34	33	29	25
A2. Primary balance is unchanged from 2013	36	37	38	39	38	38	38	37
A3. Permanently lower GDP growth 1/	36	37	40	41	41	42	46	61
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014–2015	36	38	42	43	43	43	44	44
B2. Primary balance is at historical average minus one standard deviations in 2014–2015	36	37	39	39	38	38	37	34
B3. Combination of B1-B2 using one half standard deviation shocks	36	37	38	39	39	38	38	36
B4. One-time 30 percent real depreciation in 2014	36	43	44	43	42	41	37	32
B5. 10 percent of GDP increase in other debt-creating flows in 2014	36	44	45	44	43	43	40	36
PV of Debt-to-Revenue Ratio 2/								
Baseline	161	150	161	157	153	150	145	135
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	161	149	157	152	146	141	122	101
A2. Primary balance is unchanged from 2013	161	154	167	165	163	160	155	150
A3. Permanently lower GDP growth 1/	161	154	173	174	175	175	192	244
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014–2015	161	158	182	183	183	181	183	178
B2. Primary balance is at historical average minus one standard deviations in 2014–2015	161	154	167	166	163	160	153	139
B3. Combination of B1-B2 using one half standard deviation shocks	161	153	166	166	164	162	157	146
B4. One-time 30 percent real depreciation in 2014	161	177	189	184	178	172	153	129
B5. 10 percent of GDP increase in other debt-creating flows in 2014	161	181	196	189	185	179	166	144
Debt Service-to-Revenue Ratio 2/								
Baseline	14	14	15	14	14	14	13	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	14	14	15	13	11	12	11	9
A2. Primary balance is unchanged from 2013	14	14	15	14	15	15	14	14
A3. Permanently lower GDP growth 1/	14	14	15	15	15	15	17	23
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014–2015	14	14	16	15	16	16	16	17
B2. Primary balance is at historical average minus one standard deviations in 2014–2015	14	14	15	14	13	14	13	13
B3. Combination of B1-B2 using one half standard deviation shocks	14	14	15	14	12	14	14	13
B4. One-time 30 percent real depreciation in 2014	14	15	17	17	16	17	16	16
B5. 10 percent of GDP increase in other debt-creating flows in 2014	14	14	16	25	15	17	14	13

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.